Open Source Robotics

This is the concluding part of the series on open source robotics. The previous three parts introduced the Linux enthusiast to the Robot Operating System (ROS), the Player/Stage platform and other interesting aspects of robotics. This article is for the Linux newbie.

If you are new to Linux, your curiosity will far outweigh your familiarity with this open source platform. This article will give you a gentle introduction to the world of world of robotics software in Linux.

In Ubuntu 12.04, if you do a quick search for 'robot' in the Ubuntu Software Centre, the top two results you get will be GvRng and Robocode. Let's take a look at these. One would also probably find Robots, which however, is a game and not related to robotics.

GvRng

GvRng or Guido Van Robot—the name is a play on Guido Van Rossum, the creator of the Python programming language. Though it is not a robot programming platform, one can get to learn basic robot navigation, pick and drop operations, etc. The GUI is devised using Python. The avid programmer will find GvRng similar to the Logo and Karel programming languages.

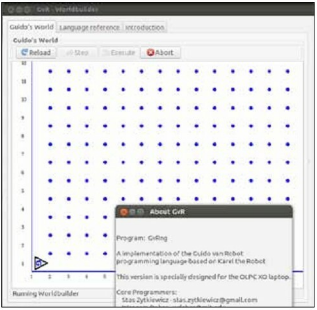

Figure 1: GvRng in action

Another similar GUI-based interface is RUR-PLE, available at *http://code.google.com/p/rur-ple/*, and it is not in Ubuntu's software repository. The developers promote it as "*…an environment designed to help you learn computer programming using the language Python.*" Coding in GvRng and RUR-PLE is very similar.

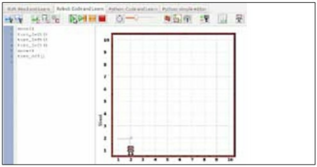

Figure 2: Code and environment in RUR-PLE

Robocode

Robocode is designed so that it appeals as a battle of tanks. However, a good lot of robot programming can be learnt from it. There is built-in code for *line following* and *obstacle avoidance.* Coded in Java, there is a Windows version of the same software.

Graduating a little, we proceed to....

Simbad, the robot simulator

It is only natural to wish for platforms that allow 3D visualisation. Probably midway between ROS and Robocode is Simbad, a Java-based 3D robot simulator with support for programming the robot in Java or Jython. It provides support for sensors, bumpers and camera. For advanced users, Simbad provides a facility to incorporate evolutionary and neural network-based algorithms into their code. However, the

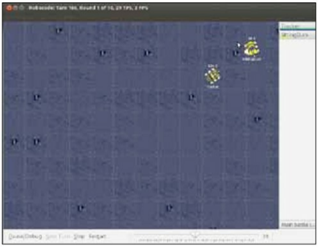

Figure 3: Battle of Tanks in Robocode